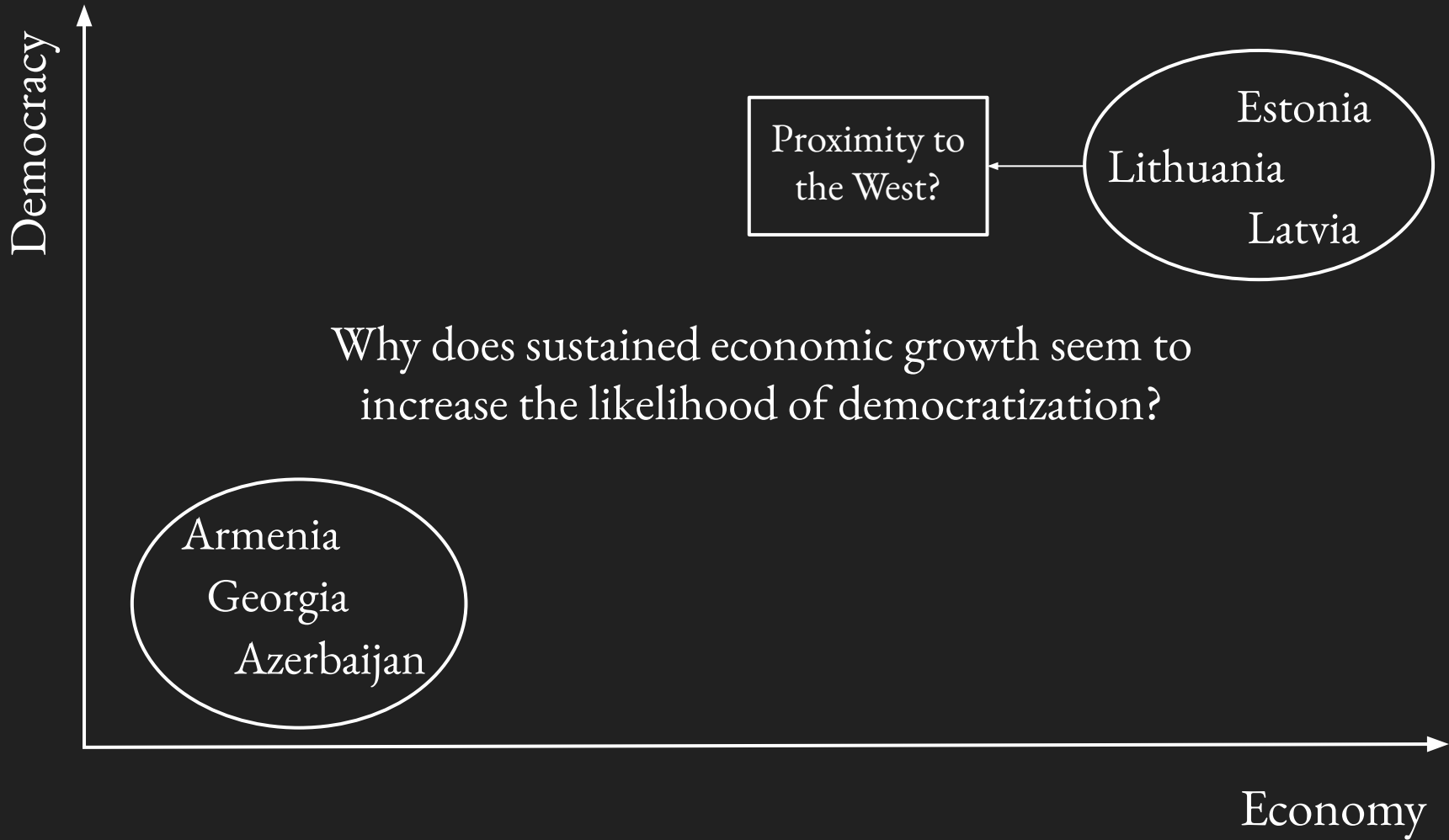


Linear Algebra for Political Scientists

Diana Kazarian
dianakazarian.github.io/laps

Structure

1. Gaussian Elimination
2. Matrix Rank
3. The Inverse
4. The Determinant



Country	Region	GDP Growth	EU Integration	Democracy Score
Armenia	South Caucasus	3	2	3
Azerbaijan	South Caucasus	4	1	2
Georgia	South Caucasus	3	3	4
Estonia	Baltics	7	9	8
Latvia	Baltics	6	8	7
Lithuania	Baltics	7	8	8

Creating an Index

Remember our scholar of the USSR who wanted to create an index for “Western alignment” using the GDP growth and EU integration columns?

$$A = \begin{bmatrix} 3 & 2 & 3 \\ 4 & 1 & 2 \\ 3 & 3 & 4 \\ 7 & 9 & 8 \\ 6 & 8 & 7 \\ 7 & 8 & 8 \end{bmatrix}$$

GDP	EU	Democracy
Growth	Integration	Score

$$\begin{bmatrix} 3 & 2 \\ 4 & 1 \\ 3 & 3 \\ 7 & 9 \\ 6 & 8 \\ 7 & 8 \end{bmatrix} \quad \begin{bmatrix} 2 \\ 1 \end{bmatrix} \quad \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

first set of weights second set of weights

But who is to say that growth should count twice as heavily as integration?

Or that they should be equally important, for that matter?

Key idea: These weights should come from the *data themselves*.

Systems of Linear Equations

Gaussian elimination is the name of the process that allows us to solve systems like these.

Estonia: $7w_1 + 9w_2 = 8$

Latvia: $6w_1 + 8w_2 = 7$



$$\begin{bmatrix} 7 & 9 \\ 6 & 8 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 8 \\ 7 \end{bmatrix}$$

		GDP Growth	EU Integration	Democracy Score
Estonia	Baltics	7	9	8
Latvia	Baltics	6	8	7

Some special (linear) combination of GDP Growth and EU Integration should reproduce these exact Democracy Score values from our dataset.

Elementary Row Operations

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} p \\ q \end{bmatrix}$$

These operations can be applied to a system of equations without changing the solution:

1. Swap two rows.

$$\left[\begin{array}{cc|c} a & b & p \\ c & d & q \end{array} \right] \begin{array}{l} R_1 \\ R_2 \end{array} \quad R_1 \leftrightarrow R_2 \quad \left[\begin{array}{cc|c} c & d & q \\ a & b & p \end{array} \right]$$

2. Multiply a row by a non-zero constant.

$$\left[\begin{array}{cc|c} a & b & p \\ c & d & q \end{array} \right] \begin{array}{l} R_1 \\ R_2 \end{array} \quad kR_1 \rightarrow R_1 \quad \left[\begin{array}{cc|c} ka & kb & kp \\ c & d & q \end{array} \right]$$

Elementary Row Operations (II)

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} p \\ q \end{bmatrix}$$

These operations can be applied to a system of equations without changing the solution:

3. Add a multiple of one row to another row.

$$\left[\begin{array}{cc|c} a & b & p \\ c & d & q \end{array} \right] \begin{array}{l} R_1 \\ R_2 \end{array}$$

$$R_2 - kR_1 \rightarrow R_2$$

$$\left[\begin{array}{cc|c} a & b & p \\ c - ka & d - kb & q - kp \end{array} \right]$$

Row Echelon Form

A matrix is in row echelon form when every row's leading nonzero entry (its *pivot*) sits strictly to the right of the pivot in the row above it, and any all-zero rows sit at the bottom. The shape doesn't depend on the matrix being square, or on any particular size: only on where the pivots (marked ●) fall relative to each other:

$$\begin{bmatrix} \bullet & * \\ 0 & \bullet \end{bmatrix} \quad \begin{bmatrix} \bullet & * & * \\ 0 & \bullet & * \\ 0 & 0 & \bullet \end{bmatrix} \quad \begin{bmatrix} \bullet & * & * & * \\ 0 & \bullet & * & * \\ 0 & 0 & 0 & \bullet \end{bmatrix} \quad \begin{bmatrix} \bullet & * & * \\ 0 & \bullet & * \\ 0 & 0 & 0 \end{bmatrix}$$

Check: Row Echelon Form or Not?

$$\begin{bmatrix} 2 & 4 & 1 \\ 0 & 3 & 5 \\ 0 & 0 & 7 \end{bmatrix} \quad \checkmark$$

$$\begin{bmatrix} 0 & 5 & 2 \\ 3 & 1 & 4 \\ 0 & 0 & 6 \end{bmatrix} \quad \times$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 4 & 1 \\ 0 & 0 & 0 \end{bmatrix} \quad \checkmark$$

$$\begin{bmatrix} 2 & 1 & 3 \\ 0 & 0 & 4 \\ 0 & 5 & 1 \end{bmatrix} \quad \times$$

Gaussian Elimination Example (Warm-Up)

$$x + y = 4$$

$$x - y = 2$$

$$R_2 - R_1 \rightarrow R_2$$

$$-2y = -2 \implies y = 1$$

$$x + 1 = 4 \implies x = 3$$

$$\left[\begin{array}{cc|c} 1 & 1 & 4 \\ 1 & -1 & 2 \end{array} \right] \begin{array}{l} R_1 \\ R_2 \end{array}$$

$$\left[\begin{array}{cc|c} 1 & 1 & 4 \\ 0 & -2 & -2 \end{array} \right]$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

Gaussian Elimination Example (II)

$$x + y = 5$$

$$2x + y = 8$$

$$R_2 - 2R_1 \rightarrow R_2$$

$$-y = -2 \Rightarrow y = 2$$

$$x + 2 = 5 \Rightarrow x = 3$$

$$\left[\begin{array}{cc|c} 1 & 1 & 5 \\ 2 & 1 & 8 \end{array} \right] \begin{array}{l} R_1 \\ R_2 \end{array}$$

$$\left[\begin{array}{cc|c} 1 & 1 & 5 \\ 0 & -1 & -2 \end{array} \right]$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

Gaussian Elimination Example (III)

$$\begin{bmatrix} 7 & 9 \\ 6 & 8 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 8 \\ 7 \end{bmatrix}$$

$$R_2 - \frac{6}{7}R_1 \rightarrow R_2$$

$$\frac{2}{7}w_2 = \frac{1}{7} \Rightarrow w_2 = \frac{1}{2}$$

$$7w_1 + 9\left(\frac{1}{2}\right) = 8$$

$$\Rightarrow 7w_1 = 8 - \frac{9}{2} = \frac{7}{2}$$

$$\Rightarrow w_1 = \frac{1}{2}$$

$$\left[\begin{array}{cc|c} 7 & 9 & 8 \\ 6 & 8 & 7 \end{array} \right] \begin{array}{l} R_1 \\ R_2 \end{array}$$

$$\left[\begin{array}{cc|c} 7 & 9 & 8 \\ 0 & \frac{2}{7} & \frac{1}{7} \end{array} \right]$$

$$\begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$$

→ The index simply averages GDP growth and EU integration.

Gaussian Elimination: Larger Systems

$$\begin{array}{l} x + y + z = 6 \\ 2x + y + 3z = 13 \\ x - y + z = 2 \end{array} \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & 1 & 3 & 13 \\ 1 & -1 & 1 & 2 \end{array} \right] \begin{array}{l} R_1 \\ R_2 \\ R_3 \end{array}$$

$$R_2 - 2R_1 \rightarrow R_2 \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -1 & 1 & 1 \\ 1 & -1 & 1 & 2 \end{array} \right]$$

$$R_3 - R_1 \rightarrow R_3 \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -1 & 1 & 1 \\ 0 & -2 & 0 & -4 \end{array} \right]$$

$$R_3 - 2R_2 \rightarrow R_3 \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -1 & 1 & 1 \\ 0 & 0 & -2 & -6 \end{array} \right]$$

$$\begin{aligned} -2z &= -6 \Rightarrow z = 3 \\ -y + 3 &= 1 \Rightarrow y = 2 \\ x + 2 + 3 &= 6 \Rightarrow x = 1 \end{aligned}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

When Elimination Goes “Wrong”

- All of our examples so far have worked out neatly: elimination produced a pivot in each row, and back-substitution gave a unique answer.
- Unfortunately, this doesn't always happen: precisely *what* goes wrong turns out to be exactly the information a political scientist needs about their data.
- Suppose our scholar's data source reports EU Integration twice: once on the familiar 0-10 scale already in his dataset, and again as a percentage out of 100 (the same underlying score, just rescaled by 10).
- If he isn't paying attention, he might include both versions as separate predictors in his model, alongside GDP Growth.

Case Study of the Baltic Countries

Let's take the data of Estonia, Latvia, and Lithuania and (try to) build an index using GDP growth, the 0-10 EU Integration score, and the 0-100 score:

$$\left[\begin{array}{ccc|c} 7 & 9 & 90 & 8 \\ 6 & 8 & 80 & 7 \\ 7 & 8 & 80 & 8 \end{array} \right] \xrightarrow{\text{Gaussian elimination}} \left[\begin{array}{ccc|c} 7 & 9 & 90 & 8 \\ 0 & \frac{2}{7} & \frac{20}{7} & \frac{1}{7} \\ 0 & 0 & 0 & \frac{1}{2} \end{array} \right]$$

This is a **false** statement. No choice of weights will make this correct.

$$\longleftarrow 0w_1 + 0w_2 + 0w_3 = \frac{1}{2}$$

Linear Independence

A set of vectors is linearly independent if none of them can be written as a combination (a weighted sum) of the others. Equivalently, the only way to combine them into the zero vector is to weight every one of them by zero.

The two EU integration columns are linearly **dependent**, as discussed.

$$(\text{integration}_{0-100}) = 10 \cdot (\text{integration}_{0-10})$$

This is a surprisingly common, easy-to-miss trap in practice: unit conversions, the same variable reported at two different resolutions, or a total column sitting alongside its own components will all produce exactly this symptom! So be careful!

Matrix Rank

The rank of a matrix is the number of pivots produced by Gaussian elimination: equivalently, the number of linearly independent rows (or, it turns out, equivalently the number of linearly independent columns). A matrix has *full rank* when its rank equals its number of rows (or columns, whichever is smaller).

$$\left[\begin{array}{ccc|c} 7 & 9 & 90 & 8 \\ 0 & \frac{2}{7} & \frac{20}{7} & \frac{1}{7} \\ 0 & 0 & 0 & \frac{1}{2} \end{array} \right]$$

This matrix has rank 2, not 3, regardless of which three countries are chosen or what their democracy scores happened to be because its columns can only ever span a 2-dimensional space.

Determining Rank: Example

$$\begin{array}{ccc} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 1 & 1 & 1 \end{bmatrix} & \begin{array}{l} R_2 - 2R_1 \rightarrow R_2 \\ R_3 - R_1 \rightarrow R_3 \end{array} & \downarrow \\ \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & -1 & -2 \end{bmatrix} & & \begin{array}{l} R_2 \leftrightarrow R_3 \end{array} \\ \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -2 \\ 0 & 0 & 1 \end{bmatrix} & & \end{array}$$

Three pivots (columns 1, 2, 3), one per row: this matrix has rank 3, full rank for a 3×3 matrix.

Rank and Solvability

For a system $A\mathbf{x} = \mathbf{b}$ with A square ($n \times n$):

- If A has full rank n , the system has exactly *one* solution, regardless of $\mathbf{b}$.
- If A is rank-deficient, the system either has *infinitely many* solutions (if the redundant rows all reduce to $0=0$) or *none* (if any redundant row reduces to $0=c$ for $c \neq 0$), depending entirely on $\mathbf{b}$.
- A system $A\mathbf{x} = \mathbf{b}$ is called **consistent** if at least one solution exists, and **inconsistent** if none exists.

The Inverse, via Elimination

For a square matrix A , its inverse (if it exists) is the matrix satisfying

$$A^{-1}A = I \quad \text{and} \quad AA^{-1} = I$$

If A^{-1} exists, solving $A\mathbf{x} = \mathbf{b}$ is immediate: multiply both sides on the left by A^{-1} .

$$A^{-1}A\mathbf{x} = A^{-1}\mathbf{b} \quad \implies \quad I\mathbf{x} = A^{-1}\mathbf{b} \quad \implies \quad \mathbf{x} = A^{-1}\mathbf{b}$$

This is the exact matrix analogue of solving $ax = b$ for a single number by multiplying both sides by a^{-1} . The identity matrix plays the role that the number 1 plays in ordinary algebra, just as we discussed last session.

Gauss-Jordan Elimination

To find A^{-1} for an $n \times n$ matrix : form the augmented matrix $[A \mid I]$, then apply row operations until the left block becomes I . Whatever ends up in the right block is A^{-1} .

Example of how to augment a matrix to prepare for inversion:

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix} \quad [A \mid I] = \left[\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{array} \right]$$

Determining the Inverse: Example

$$\left[\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{array} \right] \begin{array}{l} R_1 \\ R_2 \end{array}$$

Step 1: $R_2 - 2R_1 \rightarrow R_2$

$$\left[\begin{array}{cc|cc} 1 & 1 & 1 & 0 \\ 0 & -1 & -2 & 1 \end{array} \right]$$

Step 2: $R_1 + R_2 \rightarrow R_1$

$$\left[\begin{array}{cc|cc} 1 & 0 & -1 & 1 \\ 0 & -1 & -2 & 1 \end{array} \right]$$

Step 3: $-R_2 \rightarrow R_2$

$$\left[\begin{array}{cc|cc} 1 & 0 & -1 & 1 \\ 0 & 1 & 2 & -1 \end{array} \right]$$

$$\parallel \\ A^{-1}$$

The Determinant (2 x 2)

Gauss-Jordan elimination tells you definitively whether A is invertible, but only after you've done the work of performing the elimination. It would be useful to have a quick check, computable directly from A 's entries, that answers the yes/no question up front.

For $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, the determinant is

$$\det(A) = ad - bc$$

Example: $A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \implies \det(A) = (2)(1) - (1)(1) = 1$



A square matrix A is invertible **if and only if** $\det(A) \neq 0$.

Implications of the Determinant

- Geometrically, $\det(A)$ measures how much A stretches or shrinks area (in 2D) or volume (in higher dimensions) when it acts on vectors.
- A determinant of zero means A collapses space into a lower dimension: squashing a 2D plane down onto a single line, for instance, which is, indeed, exactly why no inverse can exist: information about the second dimension has been destroyed, and there is no way to recover it.
- The determinant formula grows more involved for larger matrices (3×3 and up), and computing it directly becomes impractical by hand well before matrices reach the sizes political scientists actually work with.
- Keep the invertibility condition in mind: $\det A = 0 \implies A$ is **not** invertible.

